



U.S. General Services
Administration

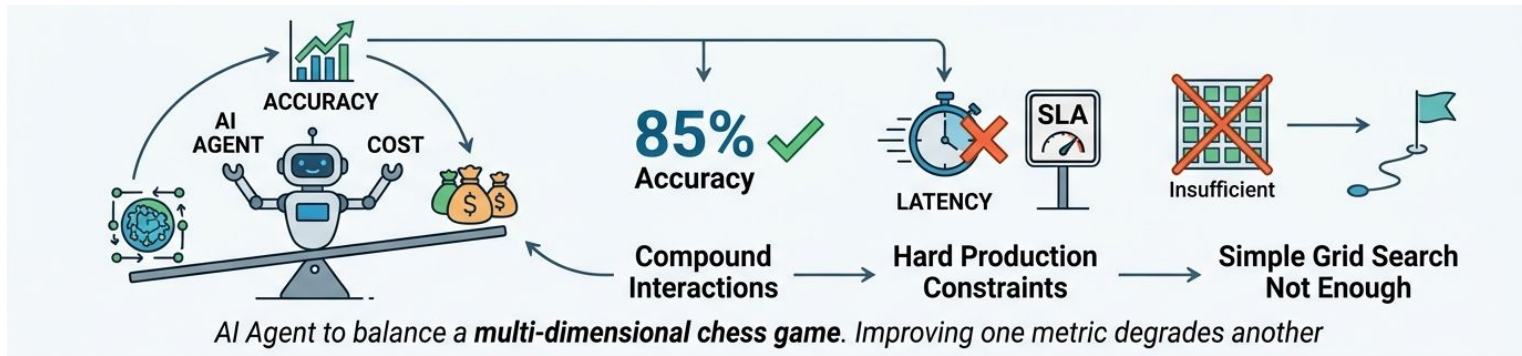


Chapter 3.4: Tuning Model Parameters for Production Performance

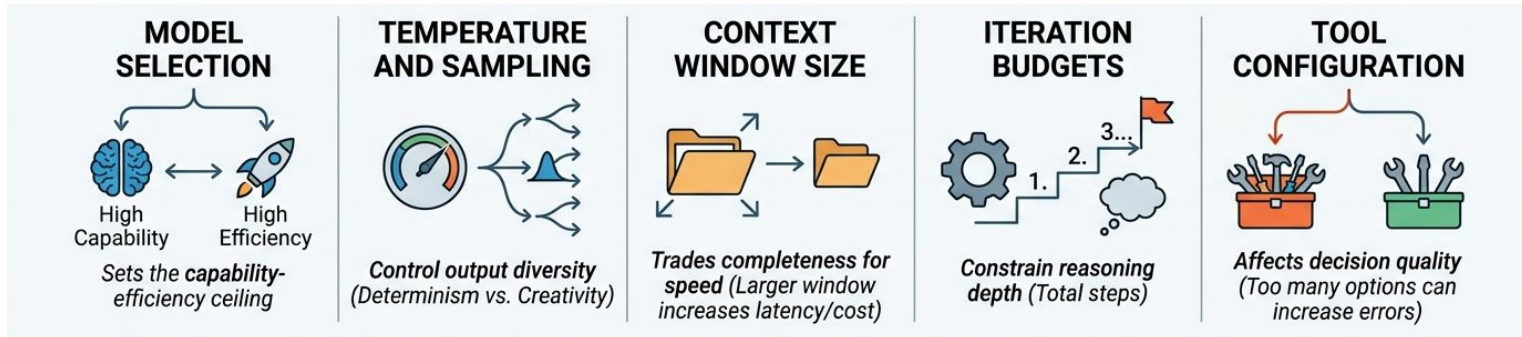
US General Service Administration

AI Community of Practice - March 2026

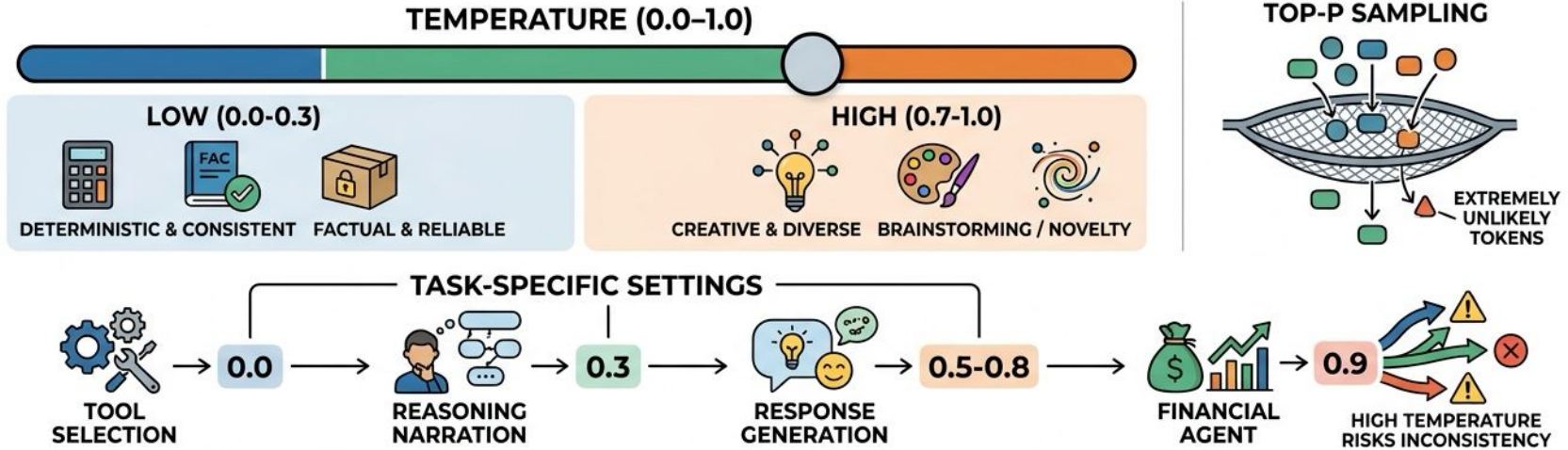
The Five Critical Parameter Categories



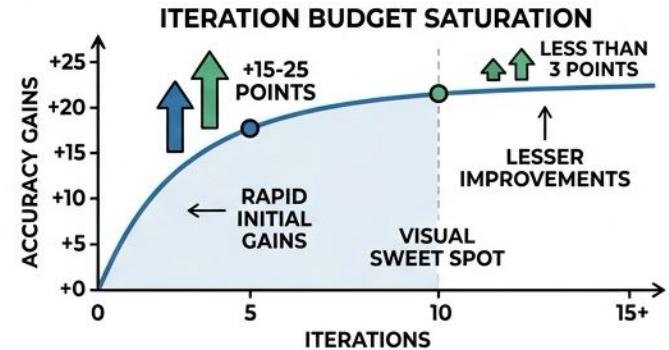
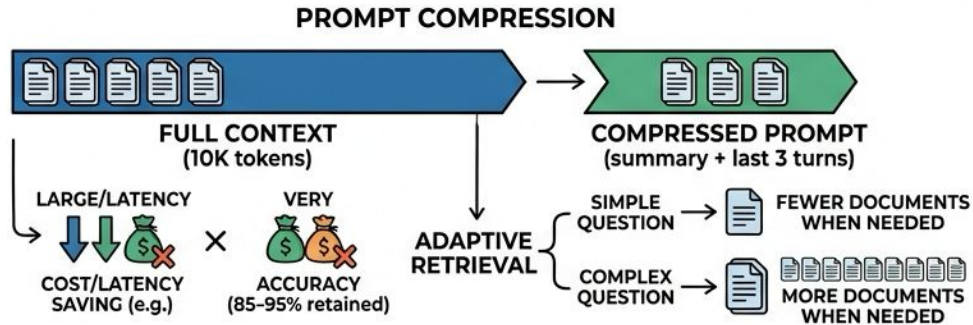
[The Five Critical Parameter Categories]



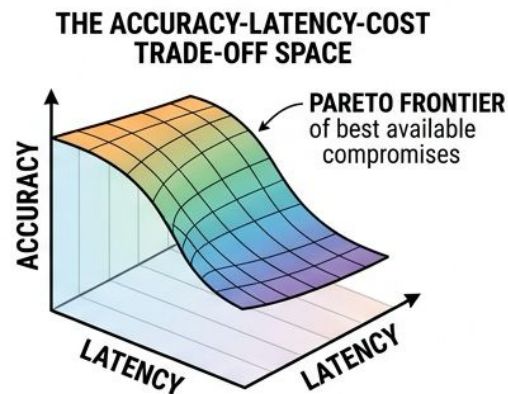
Temperature, Top-p, and Top-k Demystified



Context Windows and Iteration Budgets

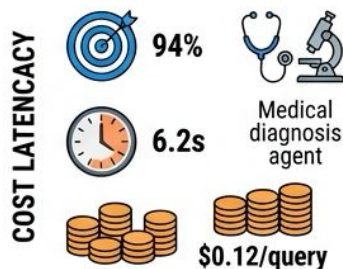


The Accuracy-Latency-Cost Trade-off Space

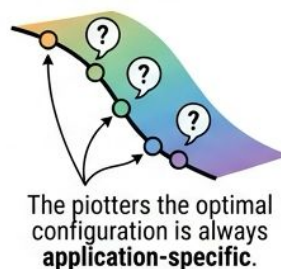


SAMPLE CONFIGURATIONS

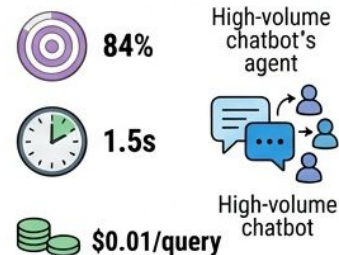
QUALITY-OPTIMIZED



PARETO-OPTIMAL CONFIGURATION

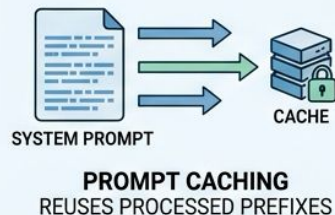
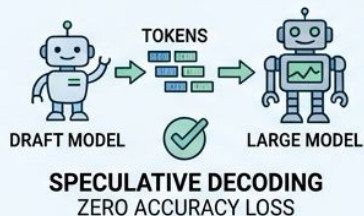
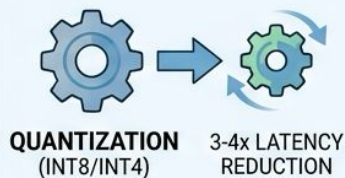
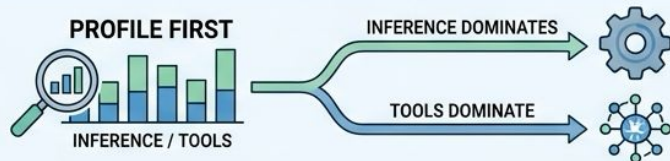


COST-OPTIMIZED

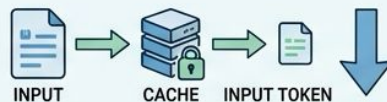


Latency and Cost Optimizations

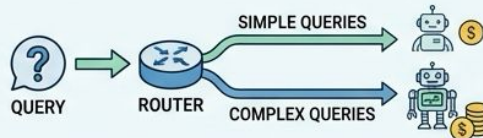
⌚ LATENCY OPTIMIZATION STRATEGIES ↓



💰 COST OPTIMIZATION -- MAKING AGENTS AFFORDABLE ↓



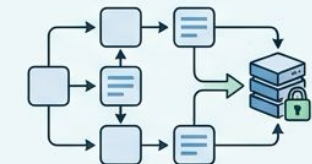
PROMPT CACHING
REDUCES INPUT TOKEN COSTS BY **60-80%**



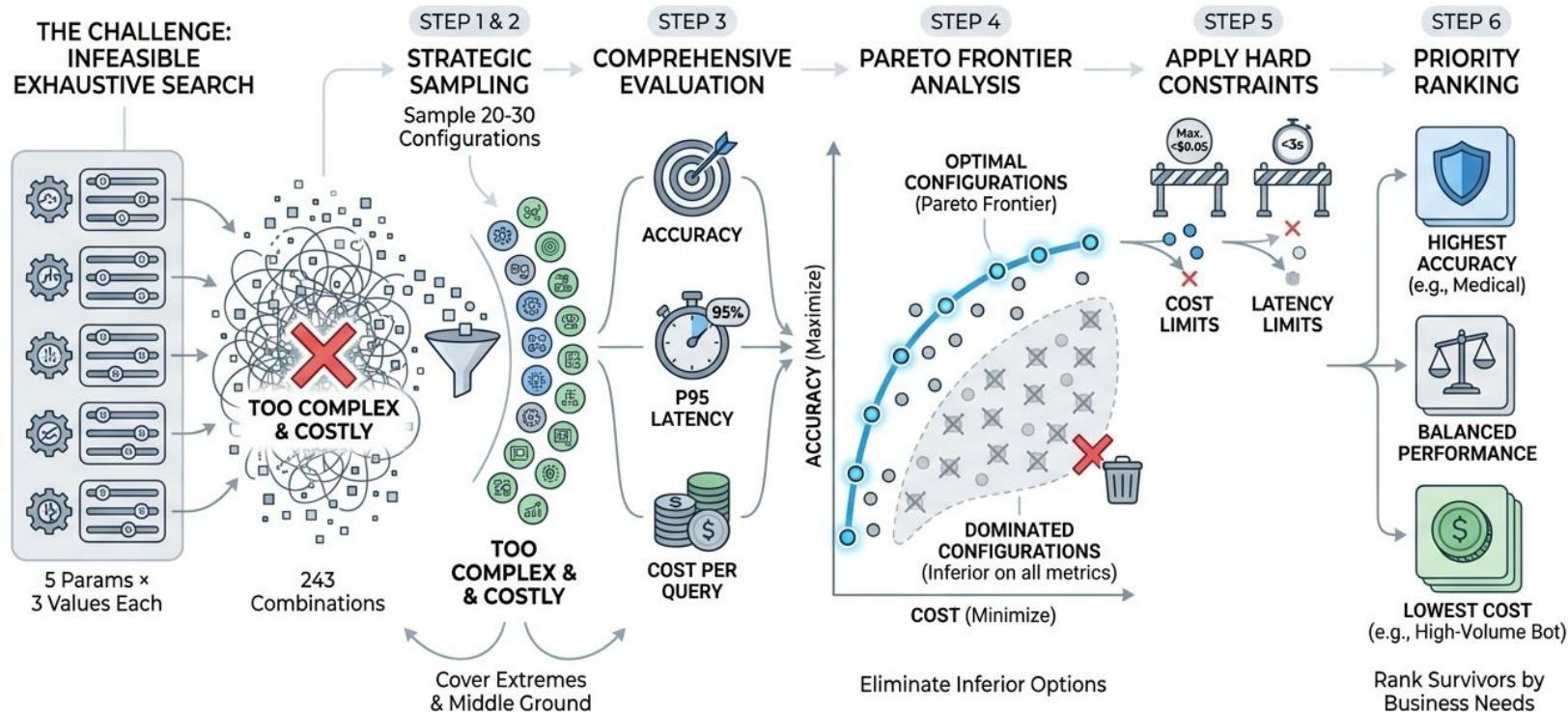
MODEL ROUTING
SENDS SIMPLE QUERIES TO CHEAP MODELS



RESPONSE CACHING
ELIMINATES COST FOR RECURRING QUESTIONS

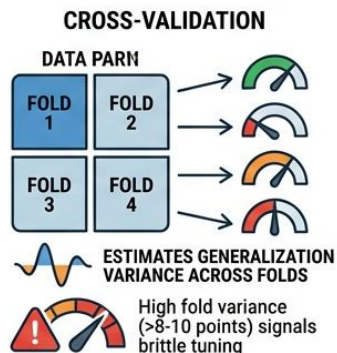
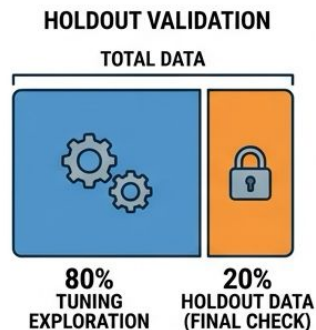
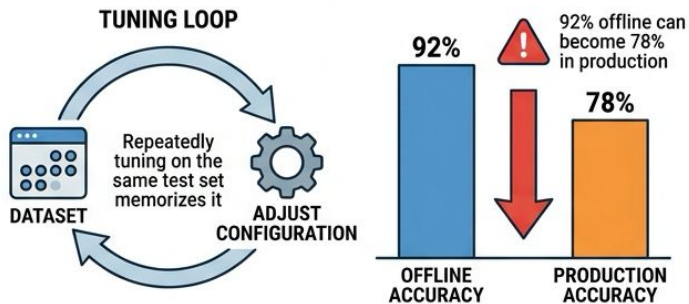


Multi-Objective Optimization and Pareto Analysis

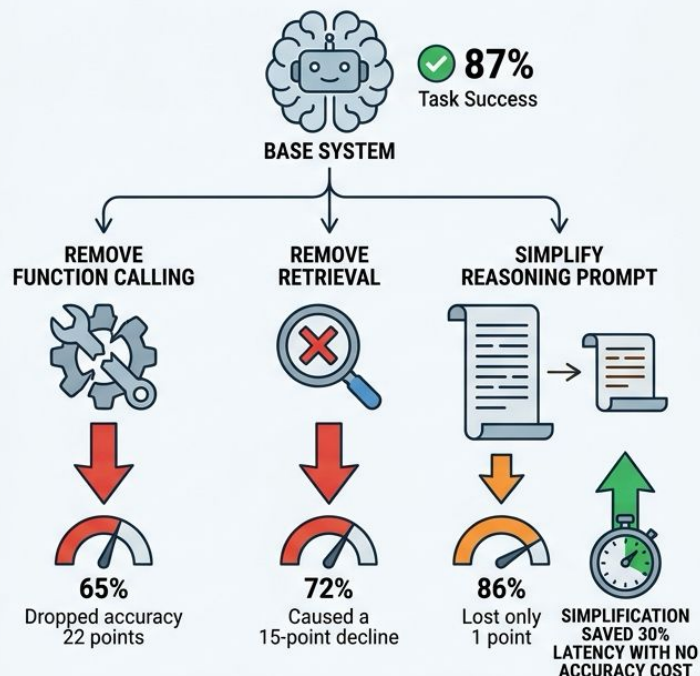


Overfitting & Ablation Studies

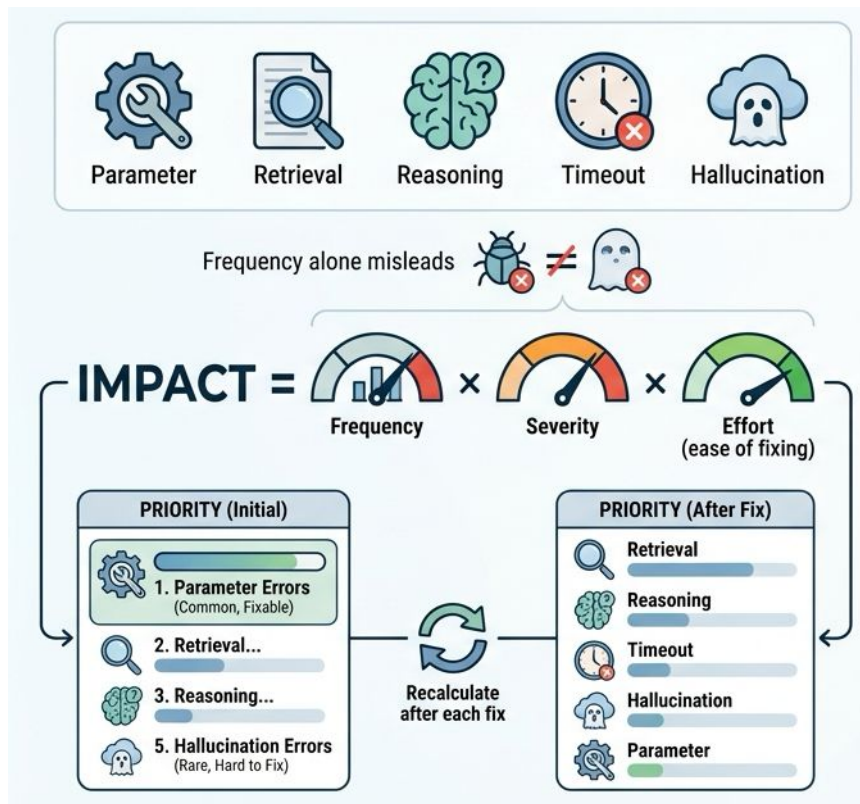
OVERFITTING -- THE SILENT KILLER OF TUNING



ABLATION STUDIES -- WHAT ACTUALLY MATTERS



Error Categorization and Impact Scoring



The Complete Improvement Cycle

